UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 20

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1 Single-Scale Covering

1.1 Review

Definition 1 (sub-Gaussian wrt d.)**.** *A process $(U_\theta)_{\theta\in\Theta}$ is **sub-Gaussian** with respect to a metric d if*

$$\mathbb{E}\exp\left(\lambda(U_\theta - U_{\theta'})\right) \le \exp\left(\frac{\lambda^2 d(\theta,\theta')^2}{2}\right) \quad \forall\lambda > 0$$

and $\mathbb{E}U_\theta = 0,\ \forall\theta,\theta' \in \Theta$.

Example: Canonical Process Let $U_\theta = \langle\theta, g\rangle$ where $g \sim N(0, I_d)$ and $\Theta \subseteq \mathbb{R}^d$.
Then

$$U_\theta - U_{\theta'} = \langle\theta - \theta', g\rangle \sim N(0, \|\theta - \theta'\|_2^2)$$

so

$$\mathbb{E}\left[e^{\lambda(U_\theta - U_{\theta'})}\right] = \exp\left(\frac{\lambda^2\|\theta-\theta'\|_2^2}{2}\right)$$

Observation. If $D := \sup_{\theta,\theta'} d(\theta,\theta')$, namely, the diameter of Θ under d.
Then we trivially have

$$\mathbb{E}\left[e^{\lambda(U_\theta - U_{\theta'})}\right] \le \exp\left(\frac{\lambda^2 D^2}{2}\right)$$

Example: Rademacher & Gaussian Complexities of B_1^d Let

$$B_1^d = \{v \in \mathbb{R}^d : \|v\|_1 \leq 1, \}, \quad \|v\|_1 = \sum_{j=1}^{d} |v_j|$$

1. Rademacher complexity is defined as

$$\mathcal{R}(B_1^d) = \mathbb{E}\left[\sup_{\theta \in B_1^d} \langle \theta, \varepsilon \rangle\right]$$

Where $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_d)$ has i.i.d. Rademacher entries. Note that:

$$\langle \theta, \varepsilon \rangle = \sum_{i=1}^{d} \theta_i \varepsilon_i \leq \sum_{i=1}^{d} |\theta_i||\varepsilon_i| = \|\theta\|_1 = 1$$

In fact, the upper-bound is attainable by choosing $\theta_i = \frac{\operatorname{sgn}(\varepsilon_i)}{d}$.
2. Gaussian complexity is defined as:

$$\mathcal{G}(B_1^d) = \mathbb{E}\left[\sup_{\theta \in B_1^d} \langle \theta, g \rangle\right], \quad g \sim N(0, I_d),$$

$$\langle \theta, g \rangle = \sum_{i=1}^{d} \theta_i g_i \leq \sum_{i=1}^{d} |\theta_i g_i| \leq \|\theta\|_1 \|g\|_\infty = \|g\|_\infty \quad \text{(by Hölder's inequality)}$$

Let $g_{i^*} = \max_i |g_i|$, then we choose θ as following to obtain the upper-bound

$$\theta_i = \begin{cases} \operatorname{sgn}(g_{i^*}), & \text{if } i = i^*, \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$\mathcal{G}(B_1^d) = \mathbb{E}\|g\|_\infty = \Theta(\sqrt{\log d})$$

Remark:

$$\mathcal{R}(B_2^d) = \sqrt{d}, \qquad \mathcal{G}(B_2^d) = \sqrt{d}(1 - o_d(1)),$$
$$\mathcal{R}(B_1^d) = 1, \qquad \mathcal{G}(B_1^d) = \sqrt{2 \log d}(1 + o_d(1)).$$

Theorem 1. *For any set T,*

$$\mathcal{R}(T) \leq \sqrt{\frac{\pi}{2}}\, \mathcal{G}(T) \leq 2\sqrt{\log d}\, \mathcal{R}(T)$$

Hint. To prove above theorem, when $g_i \sim N(0,1)$, then write

$$g_i \overset{d}{=} \varepsilon_i |g_i|, \quad \varepsilon_i \sim \text{Rad}(\frac{1}{2}), \quad \varepsilon_i \perp |g_i|$$

Then apply conditioning.

1.2 Single-Scale Covering (One-step discretization)

Theorem 2. *Suppose $\{X_\theta\}_{\theta \in T}$ is a sub-Gaussian process on the metric space (T, d). Then,*

$$\mathbb{E}\Big[\sup_{\theta \in T} X_\theta\Big] \leq \mathbb{E}\Big[\sup_{\theta,\theta' \in T} (X_\theta - X_{\theta'})\Big] \leq \underbrace{2\,\mathbb{E}\Big[\sup_{\substack{\theta,\theta' \in T \\ d(\theta,\theta') \leq \delta}} (X_\theta - X_{\theta'})\Big]}_{(A)} + 2\,\mathrm{diam}(T)\sqrt{\log \mathcal{N}(T,d,\delta)},$$

for all $\delta \in [0, \mathrm{diam}(T)]$. Here, (A) captures the local complexity.

Proof.

$$\mathbb{E}\Big[\sup_{\theta \in T} X_\theta\Big] = \mathbb{E}\Big[\sup_{\theta \in T} (X_\theta - X_{\theta'})\Big] \leq \mathbb{E}\Big[\sup_{\theta,\theta' \in T} (X_\theta - X_{\theta'})\Big]$$

The first equality comes from $\mathbb{E}[X_{\theta'}] = 0$. Fix any $\delta \in [0, \mathrm{diam}(T)]$. Take a δ-net (δ-cover) for T of size $\mathcal{N}(T, d, \delta)$. Call it Θ. For each $\theta \in T$, let $\widehat{\theta} \in \Theta$ be such that$d(\theta, \widehat{\theta}) \leq \delta$. Similarly, we define $\widehat{\theta'} \in \Theta$ such that $d(\theta', \widehat{\theta'}s) \leq \delta$. Then,

$$\begin{aligned} X_\theta - X_{\theta'} &= (X_\theta - X_{\widehat{\theta}}) + (X_{\widehat{\theta}} - X_{\widehat{\theta'}}) + (X_{\widehat{\theta'}} - X_{\theta'}) \\ &\leq 2 \sup_{\substack{\theta,\theta' \in T \\ d(\theta,\theta') \leq \delta}} (X_\theta - X_{\theta'}) + \sup_{\widehat{\theta},\widehat{\theta'} \in \Theta} (X_{\widehat{\theta}} - X_{\widehat{\theta'}}) \end{aligned}$$

Taking expectations,

$$\mathbb{E}\left[\sup_{\theta,\theta' \in T} (X_\theta - X_{\theta'})\right] \leq 2\mathbb{E}\left[\sup_{\substack{\theta,\theta' \in T \\ d(\theta,\theta') \leq \delta}} (X_\theta - X_{\theta'})\right] + \underbrace{\mathbb{E}\left[\sup_{\widehat{\theta},\widehat{\theta'} \in \Theta} (X_{\widehat{\theta}} - X_{\widehat{\theta'}})\right]}_{\text{Sup over a finite set}}$$

Since $(X_\theta - X_{\theta'})$ is a sub-Gaussian collection of $\mathcal{N}(T, d, \delta)^2$ random variables with variance proxy $\mathrm{diam}(T)^2$. For sub-Gaussian random variables X_i with variance proxy σ^2, we have

$$\mathbb{E} \max_{i=1,\ldots,M} X_i \lesssim \sigma\sqrt{2 \log M}.$$

Applying this to our collection gives

$$\mathbb{E}\left[\sup_{\widehat{\theta},\widehat{\theta}'\in\Theta}(X_{\widehat{\theta}}-X_{\widehat{\theta}'})\right]\leq \operatorname{diam}(T)\sqrt{2\log\mathcal{N}(T,d,\delta)^2}=2\operatorname{diam}(T)\sqrt{\log\mathcal{N}(T,d,\delta)}.$$

□

Example: Gaussian Process Consider the Gaussian process $\langle\theta, g\rangle$, where $g\sim N(0,I_d)$ and $\theta\in T$. And,

$$\mathcal{G}(T)=\mathbb{E}\sup_{\theta\in T}\langle\theta,g\rangle$$

Denote $D=\operatorname{diam}(T)$, then:

$$\mathcal{G}(T)\leq 2\inf_{\delta\in[0,D]}\left\{\underbrace{\mathbb{E}\left[\sup_{\substack{\theta,\theta'\in T\\ \|\theta-\theta'\|\leq\delta}}\langle\theta-\theta',g\rangle\right]}_{(*)}+D\sqrt{\log\mathcal{N}(T,\|\cdot\|,\delta)}\right\}$$

The local Gaussian complexity term (*) is of order $\delta\sqrt{d}$ since:

$$|\langle\theta-\theta',g\rangle|\leq\|\theta-\theta'\|\,\|g\|\leq\delta\,\|g\|$$

2 Chaining & Dudley's Entropy Integral

Theorem 3. *Let $u_{\theta\in\Theta}$ be a sub-Gaussian process with respect to metric d. Let D be the diameter of Θ. Then,*

$$\mathbb{E}[\sup_{\theta\in\Theta}u_\theta]\leq\inf_{\delta\in[0,D]}\{2\mathbb{E}[\sup_{d(\theta,\theta')\leq\delta}u_\theta-u_{\theta'}]+\underbrace{8\sqrt{2}\int_{\delta/4}^{D/2}\sqrt{\log\mathcal{N}(\Theta,d,\epsilon)}\,d\epsilon}_{\textit{Dudley's entropy integral}}\}$$

Remark 1. *For $\delta=0$, the upper bound is*

$$8\sqrt{2}\int_0^{D/2}\sqrt{\log\mathcal{N}(\Theta,d,\epsilon)}\,d\epsilon.$$

Proof: Start with single scale cover, make it gradually finer.

- Let Θ_j be the minimal cover for Θ at granularity $2^{-j}D$. ($|\Theta_j| = \mathcal{N}(\Theta, d, 2^{-j}D)$).
- $N \coloneqq \min\{j \geq 1 : 2^{-j}D \leq \delta\}$ ($2^{-N}D \leq \delta \leq 2^{-N+1}D$ and $\mathcal{N}(\Theta, d, 2^{-N}D) \leq \mathcal{N}(\Theta, d, \delta)$)
- *Step 1* Mimic the proof of one-step discretized

$$\mathbb{E}[\sup_{u_\theta \in \Theta} u_\theta] = 2\mathbb{E}[\sup_{d(\theta,\theta') \leq \delta} u_\theta - u_{\theta'}] + 2\mathbb{E}[\sup_{\theta_N, \theta'_N \in \Theta_N} u_{\theta_N} - u_{\theta'_N}]$$

Chaining Idea:

- $|\Theta_0| = 1$ ($\Theta_0 = \{\theta_0\}$)
- Given any θ_i, define $\theta_{i-1} \coloneqq \pi_{i-1}(\theta_i)$ as the closest element of Θ_{i-1} to θ_i.

  $$\theta_N \to \theta_{N-1} \to \ldots \to \theta_0$$

  Likewise, let

  $$\theta'_N \to \theta'_{N-1} \to \ldots \to \theta'_0 \quad (\theta'_0 = \theta_0)$$

  be the corresponding chain of θ'_N.

$$u_{\theta_N} - u_{\theta'_N} = \sum_{i=1}^{N} u_{\theta_i} - u_{\theta_{i-1}} + \sum_{i=1}^{N} u_{\theta'_{i-1}} - u_{\theta'_i} \leq \sum_{i=1}^{N} \max_{\theta_i \in \Theta_i} u_{\theta_i} - u_{\pi_{i-1}(\theta_i)} + \sum_{i=1}^{N} \max_{\theta'_i \in \Theta_i} u_{\pi_{i-1}(\theta'_i)} - u_{\theta'_i}$$

Consider $u_{\theta_i} - u_{\theta_{i-1}}$.

$$d(\theta_i, \pi_{i-1}(\theta_i)) \leq 2^{-(i-1)}D$$

So taking expectations,

$$\mathbb{E}[\max_{\theta_i \in \Theta_i} (u_{\theta_i} - u_{\theta_{i-1}})] \leq 2^{-(i-1)}D\sqrt{2\log|\Theta_i|} \leq 2^{-(i-1)}D\sqrt{2\log\mathcal{N}(\Theta, d, 2^{-i}D)}$$

Thus,

$$\begin{aligned}\mathbb{E}[\sup_{\theta \in \Theta} u_\theta] &\leq 2\mathbb{E}[\sup_{d(\theta,\theta') \leq \delta} u_\theta - u_{\theta'}] + 4\sqrt{2}\sum_{i=1}^{N} 2^{-(i-1)}D\sqrt{\log\mathcal{N}(\Theta, d, 2^{-i}D)} \\ &\leq 2\mathbb{E}[\sup_{d(\theta,\theta') \leq \delta} u_\theta - u_{\theta'}] + 8\sqrt{2}\sum_{i=1}^{N} 2^{-i}D\sqrt{\log\mathcal{N}(\Theta, d, 2^{-i}D)}\end{aligned}$$

We have

$$2^{-(i-1)}D\sqrt{\log \mathcal{N}(\Theta, d, 2^{-i}D)} \leq \int_{2^{-(i+1)}D}^{2^{-i}D} \sqrt{\log \mathcal{N}(\Theta, d, \epsilon)}\, d\epsilon.$$

Thus, we arrive at the bound

$$8\sqrt{2}\sum_{i=1}^{N} \int_{2^{-(i+1)}D}^{2^{-i}D} \sqrt{\log \mathcal{N}(\Theta, d, \epsilon)}\, d\epsilon \leq 8\sqrt{2}\int_{\delta/4}^{D/2} \sqrt{\log \mathcal{N}(\Theta, d, \epsilon)}\, d\epsilon$$

since $2^{-(N+1)}D > \delta/4$.